

CS 630: Software Testing and Quality Assurance

7-1 Paper: Code, Quality, and Strategy

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August 19, 2026

Aligning Quality Assurance Objectives with LogiCo's Organizational Goals

Quality assurance (QA) objectives should support both software quality and the larger goals of an organization. LogiCo currently faces inconsistent testing, integration problems, unreliable staging environment, and production defects. Testing is performed by developers because there is no dedicated QA team, which has resulted in different testing practices across teams. LogiCo leadership wants to reduce production defects, shorten delivery time, improve release predictability, and increase customer satisfaction. Three measurable QA objectives can help LogiCo turn these broad goals into achievable and trackable results.

Objective 1: Reduce Production Defects

LogiCo's first QA objective should be to reduce customer-impacting production defects by 40% within six months, compared with the average number of production defects during the previous three months. LogiCo should also work toward eliminating critical breaking defects that do not have a reasonable workaround.

This objective directly supports leadership's highest priority of reducing defects experienced by customers. This is particularly important because integration and contract testing are rare, and there is no consistent approach to regression testing. LogiCo's shared Oracle schema and non-versioned APIs also increase the possibility that changes to one service will create unexpected problems elsewhere on the platform.

Success would be evaluated monthly by tracking the total number of production defects, critical defects, defect severity, and defect escape rates. Comparing these measurements with the three-month baseline would show whether LogiCo is progressing toward the 40% reduction target.

Objective 2: Improve Testing and Release Efficiency

The second objective should be to reduce the median time from code merge to production-ready status by 25% within six months while maintaining at least 95% completion of required automated quality checks.

This objective supports LogiCo's goal of reducing the time required to move software into production. The company already uses Jenkins to run unit tests, build Docker containers, and perform automated API tests. However, developers sometimes skip pipeline steps when they are under time pressure, increasing the risk of inconsistent production quality.

Success would be measured using merge-to-production-ready time, automated test pass rates, pipeline completion rates, failed deployments, and rework caused by defects. LogiCo would meet the objective if delivery time decreases by at least 25% while at least 95% of releases complete required automated quality checks without increasing production defects.

Objective 3: Improve Release Predictability

The third objective should be to achieve 90% adherence to planned release dates and required QA exit criteria within two program increments. A successful release should occur within its planned release window and complete required testing without bypassing quality controls.

This objective supports LogiCo's goal of making software schedules and quality more predictable. Currently, mobile teams closely follow the SAFe schedule, while web and microservices teams frequently deploy changes outside the official release cadence. This lack of coordination contributes to instability because mobile teams depend on stable APIs and predictable releases.

Success would be evaluated by measuring the percentage of releases completed within planned windows, compliance with QA exit criteria, the number of unplanned deployments, pipeline bypasses, and post-release defects. Results should be reviewed after each sprint and program increment to determine whether the 90% target is being achieved.

Priority of the Objectives

The objectives should be prioritized as production defect reduction first, release efficiency second, and release predictability third. This order follows the priorities identified by LogiCo's leadership. Production defects have the greatest immediate impact because they affect customers and can reduce trust in the company. Release efficiency comes next because LogiCo operates under tight deadlines and needs quality improvements that do not unnecessarily delay development. Predictability follows because consistent testing and fewer defects will make releases easier to plan and manage.

The objectives also support one another. Detecting defects earlier reduces rework, which can shorten delivery time. More consistent testing and delivery practices can then make release schedules more predictable.

Strategic Alignment and Effective Software Development

Aligning QA objectives with organizational strategy helps ensure that testing provides measurable business value. Instead of simply increasing the number of tests, LogiCo can focus on outcomes that matter to customers and leadership, including fewer defects, faster delivery, and more reliable releases.

This approach is especially important because LogiCo does not have dedicated QA personnel, and improvements depend on developers adopting new practices. Leadership recognizes that changes are more likely to receive support when they reduce rework and increase confidence without significantly slowing delivery. Measurable QA objectives allow LogiCo to determine whether new practices are actually improving development outcomes and making adjustments when targets are not being met.

Conclusion

LogiCo can improve software quality by establishing measurable QA objectives that directly support its organizational priorities. Reducing production defects by 40%, improving delivery time by 25% while maintaining 95% quality-check compliance, and achieving 90% release predictability provide clear targets for evaluating progress. Prioritizing customer-impacting defects first ensures that the most urgent quality problem receives immediate attention. Together, these objectives can help LogiCo reduce rework, improve release confidence, maintain delivery speed, and provide a more reliable experience for its customers.